Incomplete Pattern — Achievers / HOTS — Answer Key

1. **B** — The dot count = (row number + column number + 1), counting from 0. Bottom-right is row 2, column 2: 2+2+1 = 5 dots.
2. **D** — The rotation = (row + column) × 90°. Bottom-right: (2+2)×90° = 360° = 0° — back to pointing straight up.
3. **C** — Sides = 3 + row + column, counting from 0. Bottom-right: 3+2+2 = 7 sides (a heptagon).
4. **A** — Every row and column has a triangle, a square, and a pentagon exactly once. Row 3 already has a pentagon and a triangle, so the missing cell must be the square.
5. **D** — Every row and column has an up-arrow, a right-arrow, and a down-arrow exactly once. Row 3 already has right and down, so the missing cell must point up.
6. **B** — Every row and column has an empty, a half-shaded, and a fully-shaded square exactly once. The missing cell must be fully shaded.
7. **C** — Size = 0.3 + 0.15 × (row + column). Bottom-right: 0.3 + 0.15×4 = 0.9 — the largest square shown yet.
8. **D** — Line count = 1 + row + column. Bottom-right: 1+2+2 = 5 lines.
9. **B** — Dot count = row + column + 2. This time the missing cell is the CENTER one (row 1, column 1): 1+1+2 = 4 dots. Use the rows and columns around it to work it out.
10. **D** — Every row and column has an up-arrow, a right-arrow, and a down-arrow exactly once. Row 1 already has up (left) and down (right), so the missing cell must point right. The middle column already has down (row 2) and up (row 3), confirming the same answer: right.
11. **C** — Sides = 3 + row + column. The missing cell is row 1, column 2: 3+1+2 = 6 sides (a hexagon). Row 2 below it (3+2+2=7, one more side) confirms the pattern.
12. **B** — Every row and column has a triangle, a square, and a pentagon exactly once. Row 2 already has a pentagon and a triangle, so the missing cell must be the square.
13. **C** — Every row and column has an empty, a half-shaded, and a fully-shaded square exactly once. Row 1 already has a fully-shaded square (middle) and a empty square (right), so the missing cell must be half-shaded.
14. **B** — Size = 0.3 + 0.15 × (row + column). The missing cell is row 2, column 0: 0.3+0.15×2 = 0.6.
15. **B** — Line count = 1 + row + column. The missing cell is row 0, column 2: 1+0+2 = 3 lines.

How to use this: 10+/15 is strong at this difficulty. For the Latin-square questions, list which figures are already in the missing cell's row AND its column — the one left out of both is your answer.